Resuscitation 85 (2014) 637–643

Contents lists available at ScienceDirect

Resuscitation

journal homepage: www.elsevier.com/locate/resuscitation

Clinical Paper

Can thoracic impedance monitor the depth of chest compressions during out-of-hospital cardiopulmonary resuscitation?☆

Erik Alonso[a,*], Digna González-Otero[a], Elisabete Aramendi[a], Sofía Ruiz de Gauna[a], Jesús Ruiz[a], Unai Ayala[a], James K. Russell[b], Mohamud Daya[c]

[a] Communications Engineering Department, University of the Basque Country UPV/EHU, Alameda Urquijo S/N, 48013 Bilbao, Spain
[b] Philips Healthcare, Bothell, WA 98021, United States
[c] Department of Emergency Medicine, Oregon Health & Science University, Portland, OR 97239-3098, United States

ARTICLE INFO

Article history:
Received 12 November 2013
Received in revised form 9 December 2013
Accepted 30 December 2013

Keywords:
Cardiac arrest
Thoracic impedance (TI)
Compression depth (CD)
Chest compression (CC)
Automated external defibrillator (AED)
Cardiopulmonary resuscitation (CPR)

ABSTRACT

Aim: To analyze the relationship between the depth of the chest compressions and the fluctuation caused in the thoracic impedance (TI) signal in out-of-hospital cardiac arrest (OHCA). The ultimate goal was to evaluate whether it is possible to identify compressions with inadequate depth using information of the TI waveform.
Methods: 60 OHCA episodes were extracted, one per patient, containing both compression depth (CD) and TI signals. Every 5 s the mean value of the maxima of the CD, D_{max}, and three features characterizing the fluctuations caused by the compressions in the TI waveform (peak-to-peak amplitude, area and curve length) were computed. The linear relationship between D_{max} and the TI features was tested using Pearson correlation coefficient (r) and univariate linear regression for the whole population, for each patient independently, and for series of compressions provided by a single rescuer. The power of the three TI features to classify each 5 s-epoch as shallow/non-shallow was evaluated in terms of area under the curve, sensitivity and specificity.
Results: The r was 0.34, 0.36 and 0.37 for peak-to-peak amplitude, area and curve length respectively when the whole population was analyzed. Within patients the median r was 0.40, 0.43 and 0.47, respectively. The analysis of the series of compressions yielded a median r of 0.81 between D_{max} and the peak-to-peak amplitude, but it decreased to 0.47 when all the series were considered jointly. The classifier based on the TI features showed 90.0%/37.1% and 86.2%/43.5% sensitivity/specificity values, and an area under the curve of 0.75 and 0.71 for the training and test set respectively.
Conclusion: Low linearity between CD and TI was noted in OHCA episodes involving multiple rescuers. Our findings suggest that TI is unreliable as a predictor of D_{max} and inaccurate in detecting shallow compressions.

1. Introduction

High quality chest compressions (CCs), i.e. compressions given with an adequate rate, depth, and full chest recoil, are crucial for the restoration of spontaneous circulation during cardiopulmonary resuscitation (CPR).[1] Several studies have shown that increasing compression force or depth during CPR increases blood flow in animals[2–4] and end-tidal CO_2 levels in humans.[5] Moreover, in out-of-hospital cardiac arrest (OHCA), increasing compression depth (CD) is associated with higher chance of survival to hospital discharge.[6–8] Current resuscitation guidelines recommend applying CCs with the force necessary to depress the sternum at least 50 mm.[9]

Shallow CCs are common both in hospital and out-of-hospital cardiac arrest.[6,7,10,11] Rescuer fatigue causes a decrease in depth over time, although the CC-rate remains constant. To overcome this problem, several mechanisms based on additional sensors have been developed for real-time CPR feedback, some of which have been integrated into automated external defibrillators (AEDs). Although they have been shown to be effective,[6] they require accessory devices.

The thoracic impedance (TI), acquired through the defibrillation pads of an AED, has been suggested as an alternative method that could be used to provide CPR feedback. This signal is recorded by injecting an alternating current between electrodes and measuring the resulting voltage. Fluctuations induced in the TI waveform by

☆ A Spanish translated version of the abstract of this article appears as Appendix in the final online version at http://dx.doi.org/10.1016/j.resuscitation.2013.12.035.

* Corresponding author.
E-mail address: erik_alonso@ehu.es (E. Alonso).

0300-9572/$ – see front matter © 2014 Elsevier Ireland Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.resuscitation.2013.12.035

CCs have been successfully used to mark the instants of CCs,[12,13] to measure CC-rate,[13,14] and to detect CC interruptions.[15,16] Moreover, manual/automatic review of episodes for quality evaluation in terms of CC-rate and CC-fraction is possible based on the TI.[12] However, the relationship between the TI fluctuations and the CD has not been established yet.

A recent study by Zhang et al.[17] reported strong correlation between the TI amplitude change and both the CD and the coronary perfusion pressure in a porcine model of cardiac arrest. In a controlled scenario with 2 rescuers and 14 similar animals, they concluded that TI could serve as a surrogate marker of CD. Contradictory results have been reported for humans in short communications. Brody et al.[18] reported greater TI fluctuations for stronger CCs, while Aramendi et al.[14] showed low correlation between CD and TI amplitude fluctuations. However, limited details of the analytical methods as well as database used were published in these two studies. The main purpose of this study was to analyze the relationship between features of TI fluctuations and CD in episodes recorded during OHCA, where multiple rescuers were involved in the resuscitation effort. Linear relationship was first evaluated for the whole population, then, within patients, and finally, for series where one patient and a single rescuer were involved. The discriminative power of the features of the TI was also evaluated. The final question to be answered is whether there is information in the TI waveform that permits identifying CCs with inadequate depth in OHCA.

2. Materials and methods

2.1. Data materials

The dataset used in this study was a subset of a large cardiac arrest registry containing 623 OHCA episodes acquired by the Tualatin Valley Fire & Rescue service (Tigard, Oregon, USA). The episodes, one per patient, were collected using the Philips HeartStart MRx monitor/defibrillator between 2006 and 2009. From the original dataset only 189 episodes had both the CD and the TI signals, and we extracted the 60 having both signals uninterruptedly recorded along the whole episode and a minimum of 1000 CCs. The CD signal (sampling rate 250 Hz) was computed from the force and the acceleration recorded through a CPR assist pad. The TI signal was recorded through the defibrillation pads by applying a sinusoidal excitation current (32 kHz, 3 mA peak-to-peak) with a resolution of 0.74 mΩ per least significant bit, a bandwidth of 0–80 Hz and a sampling rate of 200 Hz. For the analysis, isolated CCs that were not part of a group with at least 10 CCs were discarded, so a total of 116,679 CCs were analyzed.

Every episode of the dataset reflected the resuscitative efforts of multiple rescuers on the patient lying on a hard and stable surface. Crews composed of 2–6 rescuers performed 2-min CC series with stops in-between to analyze the rhythm and rotate between them. To analyze intervals where the single-rescuer–single-patient pattern was guaranteed, we defined CC series as epochs of CCs performed without interruptions by a single rescuer on a patient. Interruptions in CCs longer than 1.5 s were identified as a possible change of rescuer.[19] Only series with a minimum duration of 2 min were considered for the analysis.

2.2. Signal processing and feature extraction

The CD and TI signals were processed to compute the maximum depth of each CC and the features of the fluctuation caused by the CC in the TI waveform. First, the CD signal was processed applying a negative peak detector with a static threshold of −15 mm to automatically detect the instant of maximum depth in each CC. The depth of the CC in that instant is denoted as $D_{\mathrm{max}i}$. Taking into account that instant, the cycle of each CC was identified in the CD signal, as shown in Fig. 1, where every CC cycle is depicted as a gray band. Then, the TI signal was downsampled to 100 Hz and 0.6–7 Hz band-pass filtered in order to remove baseline drift, fluctuations caused by ventilations, and high frequency noise. The processed TI signal during the ith CC cycle was denoted as $z_{\mathrm{p}i}[n]$, where n is the time sample number.

To characterize the fluctuations induced by CCs in the TI signal, three different waveform features were computed for each CC:

- **Peak-to-peak amplitude, $Z_{\mathrm{pp}i}$:** The peak-to-trough fall in $z_{\mathrm{p}i}[n]$ as shown in Fig. 1.
- **Area, A_i:** Area of $z_{\mathrm{p}i}[n]$ as the sum of the absolute values of the samples of the signal.

$$A_i = \sum_{n=1}^{N} \left| z_{\mathrm{p}i}[n] \right| \tag{1}$$

where N is the length in samples of $z_{\mathrm{p}i}[n]$.
- **Curve length, C_i:** Length of the curve of $z_{\mathrm{p}i}[n]$, computed as follows:

$$C_i = \sum_{n=2}^{N} \sqrt{\left(\frac{1}{f_s}\right)^2 + (z_{\mathrm{p}i}[n] - z_{\mathrm{p}i}[n-1])^2} \tag{2}$$

where f_s is the sampling rate of the TI signal. This feature sums the lengths of the curve between every 2 samples.

Fig. 1 illustrates two examples where the extracted features are depicted on the CD signal (top) and on the processed TI signal (bottom). Panel a shows an example where the TI fluctuations caused by CCs are quite sinusoidal, while in panel b the fluctuations are more irregular. A_i and C_i were computed because $Z_{\mathrm{pp}i}$ was not able to characterize irregular TI fluctuations. Examples of Fig. 1 show similar $Z_{\mathrm{pp}i}$ but quite different C_i values.

In order to smooth the values of the features, every 5 s the mean values for the CD maxima, D_{max}, and for the three TI features (Z_{pp}, A, and C) were computed.

2.3. Analysis of the linear relationship

The linear relationship between D_{max} and the TI features was tested using Pearson correlation coefficient (r) and univariate linear regression for the whole population, for each patient independently, and for each series of compressions provided by a single rescuer. The performance of Spearman's rank correlation coefficient was very similar to r in our analyses, but r was used as it permitted comparison with previous studies.

The linear relationship between D_{max} and Z_{pp} in a single-rescuer–single-patient pattern was analyzed with two different sets of series. First, in order to guarantee a wide range of D_{max}, series with a standard deviation greater than 7 mm in the D_{max} were considered. A single series per patient was selected, the one with the largest standard deviation, in order to avoid the bias caused by different number of series from each patient. Selecting all series per patient did not change our findings significantly. A total of 9 series were analyzed. Univariate linear regression was used to predict D_{max} using Z_{pp}, and r computed independently for each series and jointly for the complete set of series. Second, aiming to replicate the procedure that Zhang et al.[17] used with animals, series with optimal and suboptimal CCs were selected. A series was considered suboptimal when at least 75% of the CCs were below 38 mm, and optimal when at least 75% of the CCs were above 50 mm. One series per patient was selected. When more than one series of the same

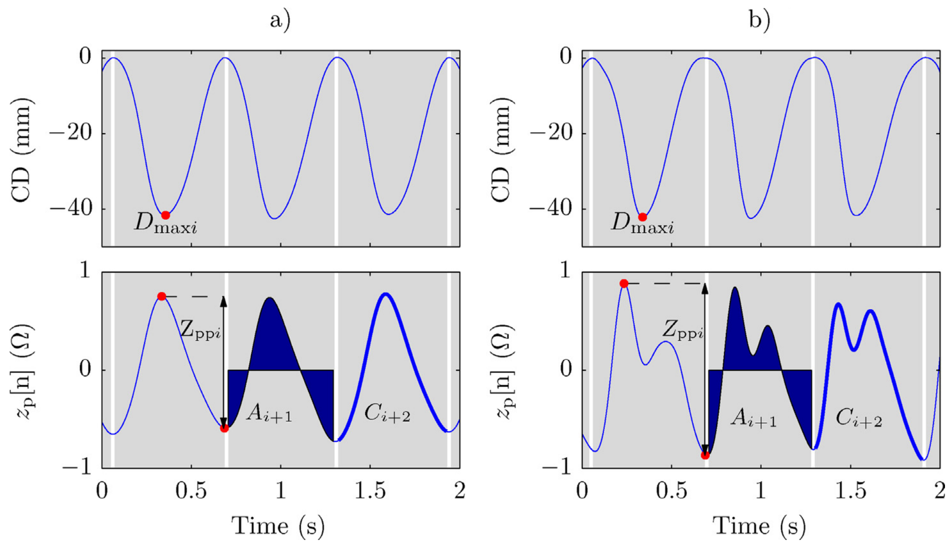

Fig. 1. Features extracted for two segments of the CD signal (upper panels) and the corresponding $z_p[n]$ signal (lower panels). For each CC cycle, represented as a gray band, the maximum depth, D_{maxi}, is represented on the CD, and the three TI features on $z_p[n]$.

patient met the criteria, the one with the largest number of CCs above (optimal group) or below (suboptimal group) the threshold was selected. A total of 12 series were jointly analyzed by computing r and fitting a model to the data through univariate linear regression.

2.4. Discrimination power of the TI features

The power of the three TI features to classify each 5 s-epoch as shallow/non-shallow was evaluated in the context of the 2005 resuscitation guidelines,[20] thinking of a possible feedback system that could be incorporated into an AED and alarm the rescuer if the compressions were shallow. According to the guidelines, CCs with $D_{max} \leq 38$ mm were classified as shallow. To enlarge the difference between shallow and non-shallow CCs, and to take into consideration measurement errors, a tolerance band of 5 mm was defined. CCs with $D_{max} \geq 43$ mm were considered as non-shallow, and those with $38 \leq D_{max} \leq 43$ mm were regarded as borderline and not considered for the evaluation of the classifier. Not considering the tolerance band would lower the discriminative power of the TI features.

A multivariate logistic regression model was applied for the classification based on a three feature vector. The regression model assigns to each 5 s-epoch a probability of non-shallow depth given by:

$$P_{\text{non-shallow}}(z) = \frac{1}{1+e^{-z}} \tag{3}$$

with $z = \theta_0 + \theta_1 \cdot Z_{pp} + \theta_2 \cdot A + \theta_3 \cdot C$

The 60 episodes were split into training (40) and test (20) sets. The regression coefficients (θ_i) were calculated for the training set using the Waikato Environment for Knowledge Analysis (WEKA) software. The discriminative power of the classifier was evaluated in terms of AUC (area under the curve), sensitivity and specificity for the training and test sets. Sensitivity was defined as the capacity to correctly identify shallow CCs and specificity as the capacity to correctly identify non-shallow CCs. The weight of the classes was set to maximize the specificity while ensuring a sensitivity above 90% in order to provide confident feedback with shallow CCs. $P_{\text{non-shallow}} > 0.5$ were classified as non-shallow.

3. Results

Results are presented as median (Q_1–Q_3), where Q_1 and Q_3 are the 1st and 3rd quartiles respectively, as their distributions did not pass the one-sample Kolmogorov–Smirnov normality test.

The 60 episodes lasted 2068 (1490–2473)s and included 1852 (1362–2556) CCs per episode. From those episodes a total of 75 series were extracted with a duration of 149 (133–187)s; every series had 277 (243–347) CCs with a depth of 42 (37–48) mm.

Fig. 2 shows the scatter-plot of D_{max} against each of the TI features for the whole population composed by 14,424 values with a depth of 42 (37–48) mm. A high dispersion was observed for the three features around the model fitted using univariate linear regression. The value of r was 0.34, 0.36, and 0.37 for Z_{pp}, A, and C, respectively.

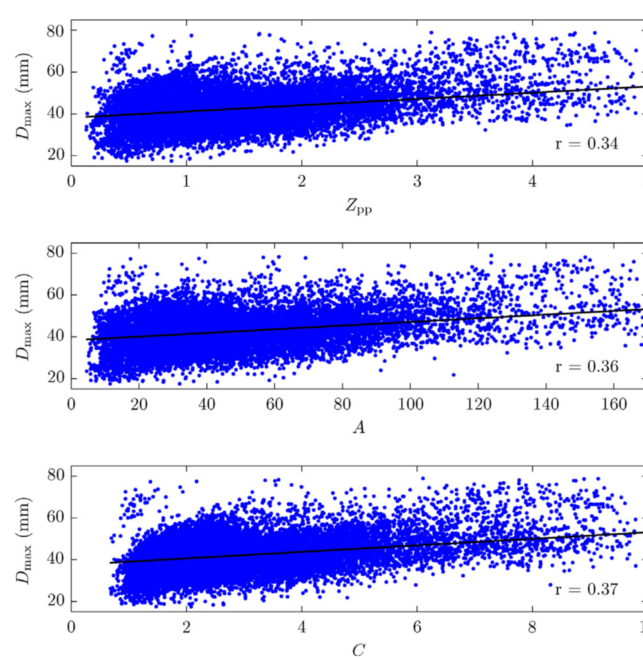

Fig. 2. Scatter-plots of the D_{max} with respect to the TI features Z_{pp}, A and C (from top to bottom respectively) for the whole population. The fitted regression line is depicted in black.

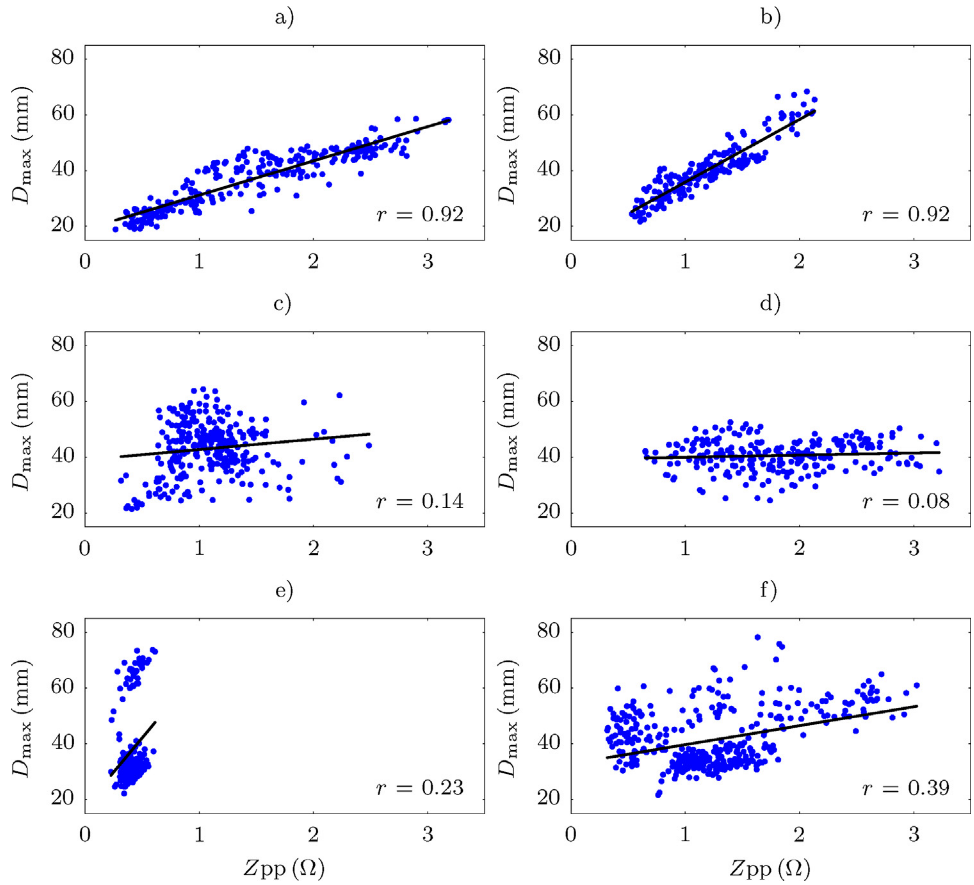

Fig. 3. Scatter-plot and linear model adjusted to the feature Z_{pp} in six patients. Panels a and b represent cases where the model fits the data well and the correlation coefficient is high. Poor fitting and poor correlation is observed for cases in panels c and d due to the high dispersion of the data. Finally, in panels e and f two different patterns may be identified.

The analysis within patients yielded a r of 0.40 (0.24–0.66) for Z_{pp}, 0.43 (0.26–0.66) for A, and 0.47 (0.25–0.68) for C. Fig. 3 shows the scatter-plot and the model fitted for the Z_{pp} in six representative patients. The cases depicted in panels a and b show a high r (0.92 for both cases), although the slopes of their regression lines differ from one patient to another. Panels c and d show two patients with a low r and a high dispersion around the regression line. Finally, in the patients depicted in panels e and f two patterns can be observed, and a single model does not adequately adjust to all the values.

Fig. 4 shows the scatter-plot for the 9 series with a minimum standard deviation of 7 mm in the D_{max}. Values for each series, 28 (26–40) per series, are depicted in a different color as well as the fitted model. The black line represents the model fitted to all the series when considered jointly (320 values with a D_{max} of 48 (40–56) mm). The individual analysis of each of the 9 series yielded a median r of 0.81 (0.51–0.83). However, when all of them were considered jointly, the r decreased to 0.47.

In Fig. 5 the scatter-plot for the set of 12 optimal and suboptimal series is presented; a total of 390 values, with a D_{max} of 57 (54–63) mm and a Z_{pp} of 3.0 (2.5–3.7) Ω for the optimal group, and a D_{max} of 32 (30–34) mm and a Z_{pp} of 0.9 (0.6–1.5) Ω for the suboptimal group. The model fitted to the data is represented as a black line, and r was 0.87.

The discrimination analysis was performed with a non-shallow group, D_{max} of 48 (45–53) mm, composed of 4426 and 1909 values corresponding to 5 s-epochs in the training and test sets respectively, and with a shallow group, D_{max} of 34 (31–36) mm, with 2773 and 1174 values in the training and test sets respectively. Wilcoxon rank sum test reported significant differences between medians

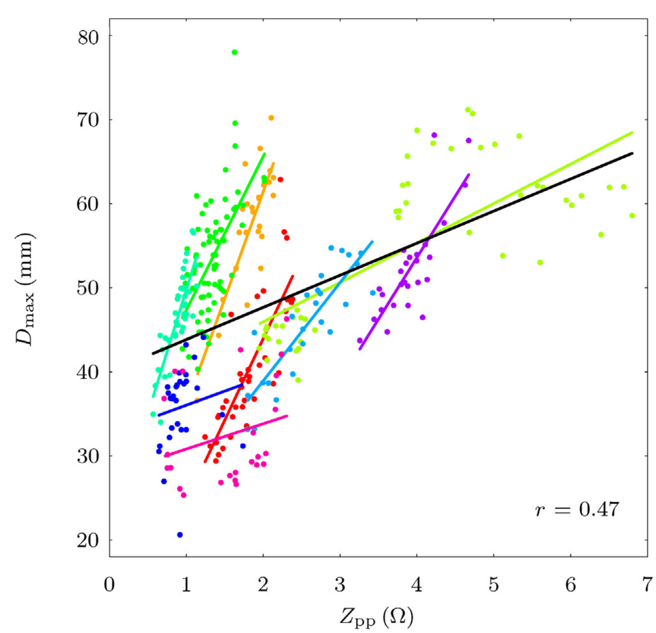

Fig. 4. Scatter-plot for the 9 series with a standard deviation greater than 7 mm in the D_{max}. The values and the regression line for each series are represented in a different color. The regression line fitted to all the data is depicted in black. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

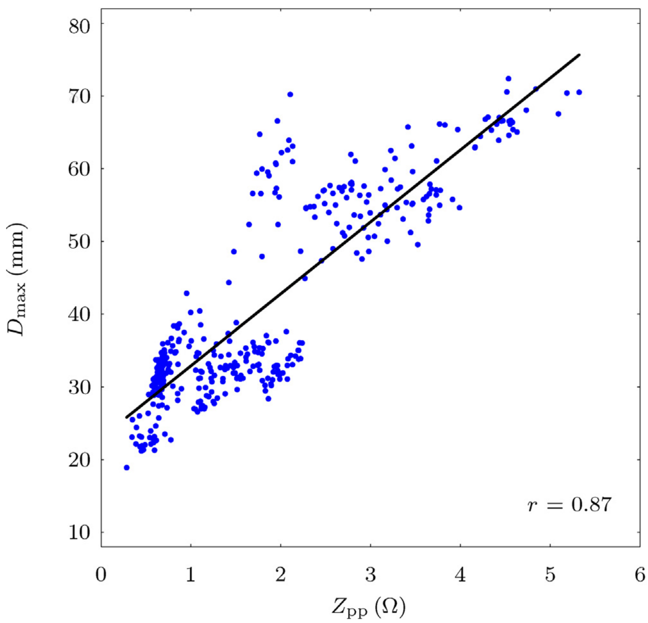

Fig. 5. Scatter-plot and regression line of D_{max} vs Z_{pp} for the set of 12 optimal and suboptimal series. The regression line is depicted in black.

for the three TI features for both groups (p-value < 0.01). The values were 1.5 (1.0–2.3) vs 1.1 (0.7– 1.5) for Z_{pp}, 50 (33–77) vs 34 (23–49) for A and 3.4 (2.2 – 5.0) vs 2.4 (1.7–3.2) for C, for non-shallow and shallow groups respectively. Fig. 6 shows the probability distribution functions for shallow (red) and non-shallow (blue) CCs. Although there were significant differences between medians, a high overlap between distributions was observed.

The logistic regression classifier showed 90.0%/37.1% sensitivity/specificity values, and AUC = 0.75 for the training set. For the test set sensitivity, specificity and AUC were 86.2%, 43.5% and 0.71, respectively.

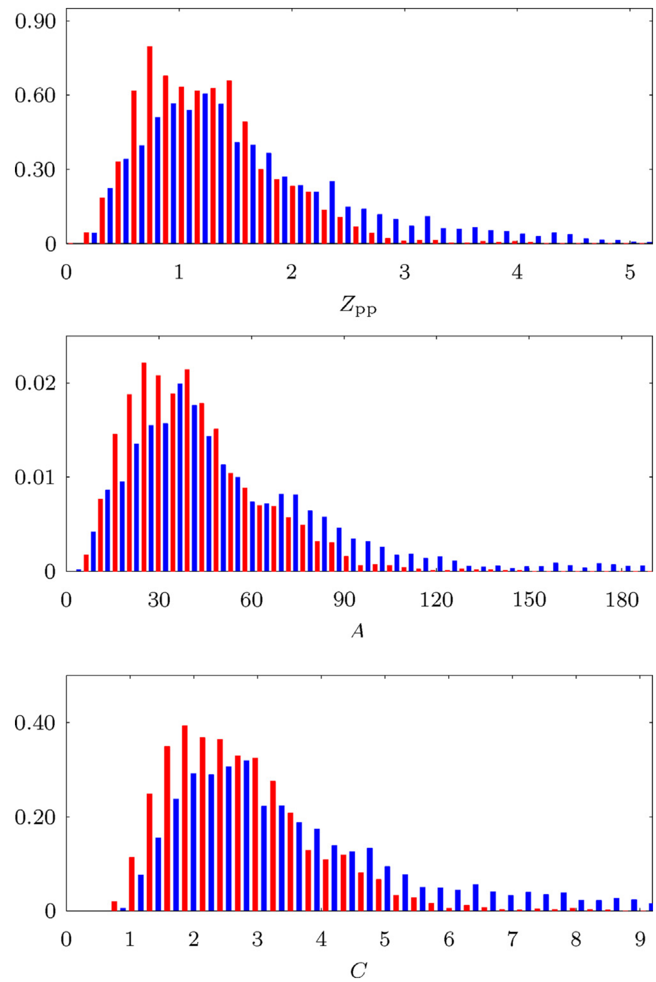

Fig. 6. Probability density functions for shallow (red) and non-shallow (blue) groups for Z_{pp}, A and C from top to bottom, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Discussion

Our study is a systematic and retrospective review of complete OHCA episodes representative of the population of patients and rescuers involved during CPR. The linear relationship between the CD and the features of the TI waveform fluctuation was analyzed, and the power of the TI to discriminate shallow CCs quantified.

In short communications, Di Maio et al.[21] and Brody et al.[18] showed that the TI reflected greater amplitude fluctuation for deeper/stronger CCs with porcine/human models respectively. Nevertheless, in none of the experiments did they have objective measurements of the CD. In 2012 Di Maio et al.[22] included information of the CD obtained from a CPR assist pad and their preliminary results for a subset of CCs showed a promising correlation ($r = 0.75$) between CD and TI fluctuations. Unfortunately, this short communication did not provide details on the analytical methods or the dataset used, so their results are not easily comparable to this study.

Our study included a set of episodes with a wide variety of patients and rescuers. A total of 60 patients, with presumably random physical characteristics, and between 2 and 6 rescuers attending each OHCA event were analyzed. We used three features to characterize the fluctuation caused by the CCs in the TI waveform (Z_{pp}, A and C). The 14,424 values obtained for each feature showed very low correlation with D_{max} ($r < 0.38$ in any case). The clouds of the three scatter-plots in Fig. 2 demonstrate that predicting D_{max} with any of the features is highly unreliable. For instance, for a Z_{pp} of 1 Ω the probability of error in the predicted value of D_{max} is high because of the wide range of corresponding D_{max} values.

The large dispersion observed in Fig. 2 was next studied by analyzing the episodes independently in order to suppress the variability introduced by different patients. The median r per episode was around 0.43 for any of the three parameters, slightly above that obtained for the whole population. However, the variability between patients was large with a r of 0.40 (0.24–0.66). Fig. 3 shows the scatter-plot of 6 distinct OHCA events where a different slope of the regression line is observed for every patient, which may reflect the individual patient-electrode characteristics. Sex, chest size, body mass and pads position have been reported to affect the baseline TI, [23–27] and TI amplitude fluctuations caused by ventilations have been shown to be correlated with the thoracic fat and the thoracic circumference.[28] Fig. 3 also shows different dispersion with respect to the regression line from one patient to another. On the one hand, with little dispersion and high slope in the regression line the r is high as can be observed in panels a and b. On the other hand, with high dispersion and low slope the correlation coefficient is low (panels c and d). Panels e and f, however, depict patients where two tendencies can be clearly distinguished. This could be linked to two distinct patterns for providing CPR by the rescuers involved in each patient. In both cases, it is clear that a single regression model would hardly fit all the values.

The impact of several rescuers was eliminated in the series analysis shown in Fig. 4, where the single-patient–single-rescuer pattern was maintained in every series. With a single rescuer the dispersion within each series clearly decreased, resulting in an increased linearity between D_{max} and Z_{pp} (0.81 (0.51–0.83)). Nevertheless, inter-patient factors such as the chest/electrodes characteristics of the 9 patients provided a low correlation, $r = 0.47$, when all the series were considered jointly, emphasizing the inability of defining a confident linear fit.

In a recent study, Zhang et al. investigated the relationship between TI changes and both CD and coronary perfusion pressure in a porcine model of cardiac arrest.[17] In 14 pigs suboptimal (35 mm) and optimal (50 mm) CCs were provided by 2 experts during 2 min. The peak-to-trough amplitude change of the TI during each CC was averaged every 5 s, obtaining a feature similar to our Z_{pp}. They found high correlation ($r = 0.89$) between their feature and D_{max}, and great differences in TI amplitude between optimal and suboptimal depth groups (1.45 ± 0.37 vs 0.47 ± 0.12, $p < 0.001$) with D_{max} (44.16 ± 4.61 vs 29.06 ± 4.90 mm $p < 0.001$).

Trying to replicate this experiment in humans, a set of 12 series with optimal and suboptimal CCs was extracted from different patients. The difference of Z_{pp} for both groups was significant (3.0 (2.5–3.7) Ω vs 0.9 (0.6–1.5) Ω, $p < 0.001$) with D_{max} (57 (54–63) mm vs 32 (30–34) mm $p < 0.001$) for optimal and suboptimal depth groups respectively. The scatter-plot obtained in Fig. 5 for D_{max} and Z_{pp} is very similar to the corresponding figure in the animal experiment. Although in both cases the correlation coefficient is high (0.89 for animals and 0.87 for humans), for a given value of Z_{pp} high variability of the predicted D_{max} was evident. For a proper interpretation of the large correlation coefficient observed, we should take into account the limitations of these analyses. On the one hand, considering only optimal and suboptimal compressions a biased picture of the OHCA episodes is shown. When the complete range of compression depths is considered, the correlation coefficient drops to 0.34 as can be observed in Fig. 2. On the other hand, the set of patients and rescuers included was small (12 patients/12 rescuers and 14 animals/2 rescuers). When a greater variety was included, 60 patients/2–6 rescuers each, a high dispersion in the scatter-plot was observed (Fig. 2), the r decreased substantially, and a given value of Z_{pp} corresponded to a wide range of D_{max}.

Although TI has been proved to be a good indicator for CPR quality parameters such as CC-rate, CC-fraction or CC interruptions,[12–16] in this study the TI has proved unreliable to predict the D_{max}. Nevertheless, from a practical perspective the power to discriminate shallow CCs was also analyzed. There is evidence that CCs with a depth of 40 mm or below reduces the percentage of patients admitted alive to hospital.[7,6] The 2005 resuscitation guidelines recommended CCs with depth in the range of 38–50 mm.[20] In our study 69% of the CCs were above the recommended minimum threshold. We tried to discriminate the group of CCs with $D_{max} \leq 38$ mm from the group with $D_{max} \geq 43$ mm. The three TI features showed distributions with different medians ($p < 0.001$) but high overlap between the two classes. The logistic regression classifier revealed an AUC = 0.71 with the test set, which implied 86.2% sensitivity for detecting shallow CCs but a 43.5% specificity. It was therefore impossible to safely identify shallow CCs using TI.

5. Conclusions

The linear relationship between CD and TI in OHCA episodes involving a wide variety of patients and multiple rescuers is poor. With our data TI cannot reliably predict D_{max} and is inaccurate in discriminating shallow CCs.

Ethical approval

CPR process information for this study was collected as part of a larger analysis looking at $EtCO_2$ in cardiac arrest. Since no identifying information was involved, the Institutional Review Board at the Oregon Health & Science University determined that this was not human research because it did not meet the definition of human subject activity per 45 Code of Federal Regulations 46.102.

Conflict of interest statement

Authors, Elisabete Aramendi, Jesús Ruiz, and Sofía Ruiz de Gauna have received research support from Bexen Cardio (Ermua, Spain) for studies on AED shock advice algorithms. James K. Russell is employed by Philips.

Acknowledgements

This work received financial support from the Ministerio de Economía y Competitividad of Spain through the projects TEC2012-31928 and TEC2012-31144, from the University of the Basque Country (UPV/EHU) through the unit UFI11/16 and from the Programa de Formación de Personal Investigador del Departamento de Educación, Universidades e Investigación del Gobierno Vasco through the grants BFI-2010-174, BFI-2010-235 and BFI-2011-166.

References

1. Meaney PA, Bobrow BJ, Mancini ME, et al. CPR quality: improving cardiac resuscitation outcomes both inside and outside the hospital a consensus statement from the American Heart Association. Circulation 2013, http://dx.doi.org/10.1161/CIR.0b013e31829d8654.
2. Maier G, Tyson G, Olsen C, et al. The physiology of external cardiac massage: high-impulse cardiopulmonary resuscitation. Circulation 1984;70:86–101.
3. Bellamy R, DeGuzman L, Pedersen D. Coronary blood flow during cardiopulmonary resuscitation in swine. Circulation 1984;69:174–80.
4. Babbs C, Voorhees W, Fitzgerald K, Holmes H, Geddes L. Relationship of blood pressure and flow during CPR to chest compression amplitude: evidence for an effective compression threshold. Ann Emerg Med 1983;12:527–32.
5. Ornato J, Levine R, Young D, Racht E, Garnett A, Gonzalez E. The effect of applied chest compression force on systemic arterial pressure and end-tidal carbon dioxide concentration during CPR in human beings. Ann Emerg Med 1989;18:732–7.
6. Kramer-Johansen J, Myklebust H, Wik L, et al. Quality of out-of-hospital cardiopulmonary resuscitation with real time automated feedback: a prospective interventional study. Resuscitation 2006;71:283–92.
7. Edelson D, Abella B, Kramer-Johansen J, et al. Effects of compression depth and pre-shock pauses predict defibrillation failure during cardiac arrest. Resuscitation 2006;71:137–45.
8. Babbs C, Kemeny A, Quan W, Freeman G. A new paradigm for human resuscitation research using intelligent devices. Resuscitation 2008;77:306–15.
9. Koster R, Baubin M, Bossaert L, et al. European Resuscitation Council guidelines for resuscitation 2010. Section 2. Adult basic life support and use of automated external defibrillators. Resuscitation 2010;81:1277–92.
10. Abella B, Alvarado J, Miklebust H, et al. Quality of cardiopulmonary resuscitation during in-hospital cardiac arrest. JAMA 2005;293:305–10.
11. Wik L, Kramer-Johansen J, Myklebust H, et al. Quality of cardiopulmonary resuscitation during out-of-hospital cardiac arrest. JAMA 2005;293:299–304.
12. Stecher F, Olsen J, Stickney R, Wik L. Transthoracic impedance used to evaluate performance of cardiopulmonary resuscitation during out of hospital cardiac arrest. Resuscitation 2008;79:432–7.
13. Aramendi E, Ayala U, Irusta U, Alonso E, Eftestøl T, Kramer-Johansen J. Suppression of the cardiopulmonary resuscitation artefacts using the instantaneous chest compression rate extracted from the thoracic impedance. Resuscitation 2012;83:692–8.
14. Aramendi E, Ayala U, Irusta U, Alonso E. Use of the transthoracic impedance to determine CPR quality parameters. Resuscitation 2010;81:S52.
15. Didon J, Krasteva V, Ménétré S, Stoyanov T, Jekova I. Shock advisory system with minimal delay triggering after end of chest compressions: accuracy and gained hands-off time. Resuscitation 2011;82(Suppl. 2):S8–15.
16. González-Otero D, de Gauna S, Ruiz J, Ayala U, Alonso E. Automatic detection of chest compression pauses using the transthoracic impedance signal. In: Computing in cardiology (CinC), 2012. 2012. p. 21–4.
17. Zhang H, Yang Z, Huang Z, et al. Transthoracic impedance for the monitoring of quality of manual chest compression during cardiopulmonary resuscitation. Resuscitation 2012;83:1281–6.
18. Brody D, Di Maio R, Crawford P, Navarro C, Anderson J. The impedance cardiogram amplitude as an indicator of cardiopulmonary resuscitation efficacy in a porcine model of cardiac arrest. J Am Coll Cardiol 2011;57:E1134.
19. Kramer-Johansen J, Edelson D, Losert H, Köhler K, Abella B. Uniform reporting of measured quality of cardiopulmonary resuscitation (CPR). Resuscitation 2007;74:406–17.

20. Handley A, Koster R, Monsieurs K, et al. European Resuscitation Council guidelines for resuscitation 2005. Section 2. Adult basic life support and use of automated external defibrillators. Resuscitation 2005;67(Suppl. 1):S7–23.
21. Di Maio R. The impedance cardiogram is an indicator of CPR effectiveness for out-of-hospital cardiac arrest victims. J Am Coll Cardiol 2010;55. A217.E2062.
22. Di Maio R, Howe A, McCanny P, et al. Is the impedance cardiogram a potential indicator of effective external cardiac massage in a human model? A study to establish if there is a linear correlation between the impedance cardiogram and depth in a cardiac arrest setting. Resuscitation 2012;83: e62.
23. Krasteva V, Matveev M, Mudrov N, Prokopova R. Transthoracic impedance study with large self-adhesive electrodes in two conventional positions for defibrillation. Physiol Meas 2006;27:1009–22.
24. Geddes L, Valentinuzzi M. Temporal changes in electrode impedance while recording the electrocardiogram with "Dry" electrodes. Ann Biomed Eng 1973;1:356–67.
25. Kerber R, Grayzel J, Hoyt R, Marcus M, Kennedy J. Transthoracic resistance in human defibrillation. Influence of body weight: chest size, serial shocks, paddle size and paddle contact pressure. Circulation 1981;63:676–82.
26. Garcia L, Kerber R. Transthoracic defibrillation: does electrode adhesive pad position alter transthoracic impedance? Resuscitation 1998;37:139–43.
27. Caterine MR, Yoerger DM, Spencer KT, Miller SG, Kerber RE. Effect of electrode position and gel-application technique on predicted transcardiac current during transthoracic defibrillation. Ann Emerg Med 1997;29:588–95.
28. Hull E, Irie T, Heemstra H, Wildevuur R. Transthoracic electrical impedance: artifacts associated with electrode movement. Resuscitation 1978;6:115–24.